TRANSFER OF COLORABILITY IN EUCLIDEAN SPACE AND NEW LOWER BOUNDS FOR THE CHROMATIC NUMBER OF $\mathbb{R}^n$

ANONYMOUS

ABSTRACT. OpenAI's preprint *The Euclidean plane is not five-colorable* (September 2026) proves, as one of its two main steps, that the plane admits a proper k-coloring (with arbitrary color classes) if and only if it admits a *weak measurable* k-coloring, i.e. a measurable coloring in which monochromatic unit pairs form a null set of pairs. The key input is a rigidity theorem for rotation-invariant laws on the character group of the discrete algebraic plane. We extend this transfer theorem to $\mathbb{R}^n$ for every $n \geq 2$. The obstacle in higher dimensions is that the algebraic rotation group $SO(n, \overline{\mathbb{Q}} \cap \mathbb{R})$ is not amenable, so the planar averaging cannot be imitated. We avoid it by averaging over rotations of a single coordinate plane only, applying planar rigidity to *twisted* push-forwards of spectral measures, and removing the remaining spectrum (continuous along the plane but wild in the complementary directions) by an invariant-mean argument combined with positivity. As a consequence, $\chi(\mathbb{R}^n) \geq \lceil 1/m_1(\mathbb{R}^n) \rceil$, where $m_1(\mathbb{R}^n)$ is the largest upper density of a measurable set avoiding distance one. Known upper bounds for m_1 then give, for example, $\chi(\mathbb{R}^3) \geq 7$ (previously 6), $\chi(\mathbb{R}^4) \geq 11$, $\chi(\mathbb{R}^8) \geq 53$, and $\chi(\mathbb{R}^n) \geq (1.268 + o(1))^n$, improving Raigorodskii's $(1.239\ldots + o(1))^n$. The planar rigidity theorem is used as a black box; all other steps are proved in full, modulo standard spectral theory and the de Bruijn–Erdős compactness theorem. The measurable bounds used are due to Oliveira–Vallentin, Bachoc–Passuello–Thiery and DeCorte–Oliveira–Vallentin; what is new is their transfer to arbitrary colorings.

1. INTRODUCTION

A *proper k-coloring* of $\mathbb{R}^n$ is a map $c : \mathbb{R}^n \to [k] = \{1, \ldots, k\}$ with $c(x) \neq c(y)$ whenever $|x - y| = 1$; $\chi(\mathbb{R}^n)$ is the least such k, no regularity being imposed. The measurable chromatic number $\chi_m(\mathbb{R}^n)$ is defined by requiring measurable color classes. Measurable colorings are accessible to Fourier-analytic methods: if $A \subset \mathbb{R}^n$ is measurable and contains no two points at distance 1, linear and semidefinite programming bound its upper density. For measurable $A \subset \mathbb{R}^n$ define the *centred upper density* and the *uniform upper density*

$$\overline{\delta}(A) = \limsup_{T \to \infty} \frac{|A \cap [-T, T]^n|}{(2T)^n}, \qquad \overline{\delta}^*(A) = \sup_{p \in \mathbb{R}^n} \limsup_{T \to \infty} \frac{|A \cap (p + [-T, T]^n)|}{(2T)^n}.$$

Taking $p = 0$ shows $\overline{\delta}(A) \leq \overline{\delta}^*(A)$. Let

$$m_1(\mathbb{R}^n) = \sup\{\overline{\delta}(A) : A \text{ measurable}, |a - a'| \neq 1 \ \forall a, a' \in A\}, \tag{1}$$

and let $m_1^*(\mathbb{R}^n)$ be defined in the same way with $\overline{\delta}^*$. Then $m_1(\mathbb{R}^n) \leq m_1^*(\mathbb{R}^n)$, so every upper bound for m_1^* is an upper bound for m_1. Oliveira–Vallentin [13] and Bachoc–Passuello–Thiery [1] use the centred density $\overline{\delta}$ (cubes centred at the origin). DeCorte–Oliveira–Vallentin [6] use the uniform density $\overline{\delta}^*$ (supremum over all centres). Since some color class of a measurable k-coloring has centred upper density at least $1/k$, one has $\chi_m(\mathbb{R}^n) \geq 1/m_1(\mathbb{R}^n)$. This route, developed by Oliveira and Vallentin [13], Bachoc, Passuello and Thiery [1], and DeCorte, Oliveira and Vallentin [6], gives bounds for $\chi_m(\mathbb{R}^n)$ that exceed all lower bounds for $\chi(\mathbb{R}^n)$ known to us in every dimension $n \geq 3$. For $n = 3$ this already holds with the 2010 bound of [13] ($7 > 6$). For

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$n = 4$ it requires the bound of [6] ($11 > 9$), since [13] and [1] give 9 and 10 respectively. Until recently there was no way to transfer them to arbitrary colorings.

In September 2026 OpenAI released a preprint [12] proving $\chi(\mathbb{R}^2) \geq 6$. One of its two main steps is a *transfer theorem* [12, Theorem 1.3]: for every k, the plane has a proper k-coloring if and only if it has a *weak measurable k-coloring* (Definition 1.1 below with $n = 2$). The heart of the transfer is a rigidity theorem [12, Theorem 2.3]: every probability measure on the character group of the discrete group $\mathbb{F}(i)$ ($\mathbb{F}$ the real algebraic numbers) that is invariant under the algebraic rotations and gives no mass to the ordinary continuous characters is Haar measure. According to its accompanying repository (file `lean/docs/158.md`), the planar five-color result, whose proof includes the transfer, has been formalized in Lean. We have not examined the Lean sources.

Definition 1.1 (Weak measurable coloring)**.** Let $n \geq 2$, $k \geq 1$, let σ be the normalized surface measure on S^{n-1}, and let $B(0, R)$ be the open ball of radius R. A *weak measurable k-coloring* of $\mathbb{R}^n$ is a Lebesgue measurable map $c : \mathbb{R}^n \to [k]$ with classes $A_i = c^{-1}(i)$ such that for every $R > 0$

$$\sum_{i=1}^{k} \int_{B(0,R)} \int_{S^{n-1}} \mathbf{1}_{A_i}(x)\mathbf{1}_{A_i}(x+u)\, d\sigma(u)\, dx = 0.$$

As in the plane, the condition is unchanged if c is modified on a null set (for each u, $x \mapsto x+u$ preserves Lebesgue measure; apply Fubini).

Theorem 1.2 (Transfer of colorability in $\mathbb{R}^n$)**.** *Let $n \geq 2$ and $k \geq 1$. In ZFC, $\mathbb{R}^n$ admits a proper k-coloring if and only if it admits a weak measurable k-coloring.*

For $n = 2$ this is [12, Theorem 1.3]. Our contribution is the case $n \geq 3$. Apart from the planar rigidity theorem, the proof is complete modulo standard spectral theory (the SNAG and Bochner theorems) and the de Bruijn–Erdős compactness theorem.

Corollary 1.3. *For every $n \geq 2$, $\chi(\mathbb{R}^n) \geq \lceil 1/m_1(\mathbb{R}^n) \rceil$, with m_1 defined by the centred upper density as in* (1)*. The same holds for m_1 defined with any fixed exhaustion, such as balls $B(0, T)$. Since $m_1 \leq m_1^*$, also $\chi(\mathbb{R}^n) \geq \lceil 1/m_1^*(\mathbb{R}^n) \rceil$.*

Corollary 1.4. *$\chi(\mathbb{R}^3) \geq 7$, $\chi(\mathbb{R}^4) \geq 11$, $\chi(\mathbb{R}^5) \geq 17$, $\chi(\mathbb{R}^6) \geq 23$, $\chi(\mathbb{R}^7) \geq 39$, $\chi(\mathbb{R}^8) \geq 53$, and $\chi(\mathbb{R}^n) \geq (1.268 + o(1))^n$ as $n \to \infty$. More values appear in Table 1.*

The corresponding bounds for the *measurable* chromatic number are not new: DeCorte, Oliveira and Vallentin [6, Table 2] state $\chi_m(\mathbb{R}^n) \geq 7, 11, 17, 23, 39, 53$ for $n = 3, \ldots, 8$, and the case $\chi_m(\mathbb{R}^3) \geq 7$ is due to Oliveira and Vallentin [13]. Our contribution is their transfer to χ. For unrestricted colorings, the previous best lower bound in dimension three was $\chi(\mathbb{R}^3) \geq 6$ (Nechushtan [11]); the best upper bound is 15 (Coulson [5]; Radoičić–Tóth [15]). Asymptotically, the best previous lower bound was $\chi(\mathbb{R}^n) \geq (1.239\ldots + o(1))^n$ (Raigorodskii [16]), and the best upper bound is $(3 + o(1))^n$ (Larman–Rogers [10]).

The difficulty and the idea. The planar proof [12, §4] restricts a proper coloring to the countable plane $E = \mathbb{F}(i)$, averages it over the amenable group $E \rtimes K$ (K the algebraic rotations) to get a stationary random labeling, splits each label indicator into its *continuous part* p_i (spectrum on ordinary characters) and a remainder b_i, and kills all correlations of b_i with rigidity, since the spectral measure of b_i is K-invariant and wild. The continuous parts then define a weak measurable coloring.

In $\mathbb{R}^n$, $n \geq 3$, the analogous group $\mathbb{F}^n \rtimes SO(n, \mathbb{F})$ is not amenable ($SO(3, \mathbb{F})$ contains free subgroups), so there may be no invariant law under all algebraic rotations. We average over only one planar rotation group $K = SO(2, \mathbb{F})$ acting on the first coordinate plane P. Spectrum

that is wild along P is removed by rigidity applied, for each offset y in the complementary space W, to the push-forward to $\widehat{P}$ of the spectral measure twisted by the character value $d(0, y)$. The spectrum that is continuous along P but wild along W is not shown to vanish. Instead we show that its sphere correlations vanish *after averaging over* W with an invariant mean (as in [12, Lemma 2.5]). Since the continuous part has nonnegative correlations, this forces the continuous-part correlations to vanish pointwise. This is all the measurable construction needs.

What is new and what is not. The rigidity theorem [12, Theorem 2.3] and the planar method (invariant laws, continuous factors, density points) are OpenAI's. We claim the reduction of the n-dimensional problem to the planar rigidity theorem (Sections 4–5) and the numerical consequences (Section 7). The paper [12] is a preprint; our results depend on the correctness of its Theorem 2.3.

2. Preliminaries

2.1. **Fields, groups and characters.** Let $\mathbb{F} = \overline{\mathbb{Q}} \cap \mathbb{R}$ be the field of real algebraic numbers. It is countable, dense in $\mathbb{R}$, and real closed; in particular every positive element of $\mathbb{F}$ has a square root in $\mathbb{F}$. Fix $n \geq 3$ (for $n = 2$ see Remark 5.9). Put

$$V = \mathbb{F}^n, \qquad P = \mathbb{F}^2 \times \{0\}^{n-2}, \qquad W = \{0\}^2 \times \mathbb{F}^{n-2},$$

regarded as discrete countable abelian groups, so $V = P \oplus W$. We write elements of V as (w, y) with $w \in \mathbb{F}^2$, $y \in \mathbb{F}^{n-2}$, and use the same notation for $\mathbb{R}^n = \mathbb{R}^2 \times \mathbb{R}^{n-2}$. Let $|\cdot|$ be the Euclidean norm. Let

$$K = SO(2, \mathbb{F}) = \{u \in SO(2, \mathbb{R}) : \text{entries of } u \text{ in } \mathbb{F}\},$$

acting on $\mathbb{R}^n$ by $u(x, y) = (ux, y)$. It preserves V, P and W, fixes W pointwise, and acts on V by group automorphisms. Identifying $\mathbb{F}^2$ with $E = \mathbb{F}(i)$ via $(a, b) \mapsto a + ib$, K is the group $\{u \in E : u\bar{u} = 1\}$ acting by multiplication, as in [12, Definition 2.1].

Let $D_n = \widehat{V}$ and $D_P = \widehat{P}$ be the compact, metrizable duals of the discrete groups V and P. We write $\psi_z(d) = d(z)$ for $z \in V$, $d \in D_n$. Let $\pi : D_n \to D_P$ be restriction to P; it is a continuous homomorphism. K acts on D_n and on D_P by $(u \cdot d)(z) = d(uz)$, by homeomorphic group automorphisms, and

$$\pi(u \cdot d) = u \cdot \pi(d), \qquad (u \cdot d)(0, y) = d(0, y) \qquad (u \in K,\ d \in D_n,\ y \in W). \tag{2}$$

Under the identification $P \cong E$, D_P is the group D of [12, Definition 2.1] and the K-action is the one used there.

For $m \geq 1$, a dense subgroup $\Gamma \subset \mathbb{R}^m$ and $\xi \in \mathbb{R}^m$, let $j_\Gamma(\xi) \in \widehat{\Gamma}$ be the character $z \mapsto e^{i\xi \cdot z}$. Its image is the group of *ordinary continuous characters*.

Lemma 2.1. *Let $\Gamma \subset \mathbb{R}^m$ be a countable dense subgroup. Then $j_\Gamma : \mathbb{R}^m \to \widehat{\Gamma}$ is a continuous injective homomorphism, its image C_Γ is Borel, and $j_\Gamma^{-1} : C_\Gamma \to \mathbb{R}^m$ is Borel. A character $d \in \widehat{\Gamma}$ lies in C_Γ iff d is continuous on Γ for the topology induced from $\mathbb{R}^m$.*

Proof. Continuity is pointwise continuity in ξ on each coordinate z. If $j_\Gamma(\xi) = 1$ then $z \mapsto e^{i\xi \cdot z}$ is a continuous character of $\mathbb{R}^m$ equal to 1 on the dense set Γ, hence $\equiv 1$, so $\xi = 0$. The image is the union of the compact sets $j_\Gamma(\overline{B(0, N)})$, so it is Borel ($\sigma$-compact), and the inverse is continuous on each of these compact sets, hence Borel. Finally, a character of Γ continuous in the induced topology is uniformly continuous (it is a homomorphism), so it extends to a continuous character of $\mathbb{R}^m$, which has the form $e^{i\xi \cdot x}$. □

Let $C_P \subset D_P$ and $C_n \subset D_n$ denote the ordinary continuous characters of $P \cong \mathbb{F}^2$ and $V = \mathbb{F}^n$, and put

$$A_P = \pi^{-1}(C_P) \subset D_n.$$

These sets are Borel. C_P is K-invariant: $(u \cdot j(\xi))(z) = e^{i\xi \cdot uz} = e^{i(u^T\xi)\cdot z}$. By (2), A_P is K-invariant. Since the restriction of a continuous character of V to P is continuous, $C_n \subset A_P$.

2.2. **Spectral theory.** We use the spectral theorem for unitary representations of locally compact abelian groups (Stone–Naimark–Ambrose–Godement) and Bochner's theorem, as in [2, Theorems D.3.1 and D.2.2]. If $z \mapsto T_z$ is a unitary representation of a countable discrete abelian group Γ on a separable Hilbert space H, there is a unique projection-valued Borel measure $\mathcal{E}$ on $\widehat{\Gamma}$ with $T_z = \int \psi_z \, d\mathcal{E}$. For $v \in H$, the *spectral measure* $\lambda_v(B) = \langle \mathcal{E}(B)v, v\rangle$ is the unique finite positive Borel measure with

$$\langle T_z v, v\rangle = \int_{\widehat{\Gamma}} d(z)\, d\lambda_v(d) \qquad (z \in \Gamma).$$

Uniqueness of a finite (complex) measure on a compact metrizable abelian group with given Fourier coefficients holds because the characters span a unital, conjugation-closed algebra separating points, which is uniformly dense in $C(\widehat{\Gamma})$ by Stone–Weierstrass. We use the inner product $\langle a, b\rangle = \int a\bar{b}$, linear in the first variable. Note that $\lambda_{\mathcal{E}(B)v} = \mathbf{1}_B \lambda_v$ and that $\mathcal{E}(B)$ commutes with every T_z.

Lemma 2.2. *Let $z \mapsto T_z$ be a unitary representation of V with projection-valued measure $\mathcal{E}_n$ on D_n. The projection-valued measure of the restricted representation of P is $\mathcal{E}_n \circ \pi^{-1}$. For every v, the spectral measure of v for P is $\pi_* \lambda_v$.*

Proof. $B \mapsto \mathcal{E}_n(\pi^{-1}B)$ is a projection-valued measure on D_P, and $\int \psi_w \, d(\mathcal{E}_n \circ \pi^{-1}) = \int \psi_w \circ \pi \, d\mathcal{E}_n = T_w$ for $w \in P$. Uniqueness in the spectral theorem gives the first claim, and the second follows. □

2.3. **Invariant means.** A countable abelian group, and more generally any countable group that is an extension of an abelian group by an abelian group, is amenable [2, Theorem G.2.1, Proposition G.2.2(ii)]. A continuous action of a countable amenable group on a nonempty compact metrizable space has an invariant Borel probability measure. Take a Følner sequence (Φ_N) of finite sets [2, Theorem G.5.1], average the point masses $\delta_{g\omega_0}$ over $g \in \Phi_N$, and take a weak-$*$ limit point. Every limit point is invariant by the Følner property.

3. Continuous vectors in $\mathbb{R}^m$

The next lemma is [12, Lemma 4.1] for $m = 2$, without its rotation clause, which we do not need. We need it for $m = 2$ and for $m = n$, and we give the proof for general m. Let $\Gamma = \mathbb{F}^m$ with the topology induced from $\mathbb{R}^m$. Let (Ω, μ) be a standard probability space with a measure-preserving action $(\tau_z)_{z\in\Gamma}$ of the discrete group Γ, and Koopman operators $T_z f = f \circ \tau_z$ on $L^2(\mu)$. Let $\mathcal{E}$ be the projection-valued measure on $\widehat{\Gamma}$ and $C = C_\Gamma$.

Lemma 3.1. *The closed subspace*

$$H^\Gamma = \Big\{ v \in L^2(\mu) : \lim_{z\to 0,\ z\in\Gamma} \|T_z v - v\|_2 = 0 \Big\}$$

is the range of $\mathcal{E}(C)$, and $H^\Gamma = L^2(\Omega, \mathcal{G}, \mu)$ for a complete sub-σ-algebra $\mathcal{G}$. Its orthogonal projection is the conditional expectation $\mathbb{E}(\cdot \mid \mathcal{G})$; in particular it preserves nonnegativity and fixes 1*. The operators $T_z|_{H^\Gamma}$ extend uniquely to a strongly continuous unitary representation $(T_x)_{x\in\mathbb{R}^m}$ on H^Γ, with $T_x \to T_{x_0}$ strongly as $\Gamma \ni z \to x_0$. Each T_x preserves the cone of nonnegative functions and fixes* 1*.*

Proof. Closedness and invariance. If $w \in H^\Gamma$ and $\|v-w\| \le \epsilon$, then $\|T_z v - v\| \le 2\epsilon + \|T_z w - w\|$, so H^Γ is closed. It contains constants. It is $T_{z'}$-invariant because $\|T_z T_{z'} v - T_{z'} v\| = \|T_z v - v\|$.

Spectral identification. We have $\|T_z v - v\|^2 = \int |\psi_z - 1|^2 \, d\lambda_v$. If $\lambda_v(C^c) = 0$, then writing $d = j(\xi)$, the integrand is $|e^{i\xi\cdot z} - 1|^2 \le 4$ and tends to 0 as $z \to 0$ for each ξ, so dominated convergence gives $v \in H^\Gamma$. Conversely, let $v \in H^\Gamma$ and $F(z) = \langle T_z v, v\rangle$. Then $|F(z+h) - F(z)| \le \|v\| \, \|T_h v - v\|$, so F is uniformly continuous on Γ and extends to a continuous function $\tilde{F}$ on $\mathbb{R}^m$. $\tilde{F}$ is positive definite: for $x_1, \dots, x_N \in \mathbb{R}^m$ and $c \in \mathbb{C}^N$, approximate the x_a by points of Γ and pass to the limit in $\sum c_a \bar{c}_b F(x_a - x_b) \ge 0$. By Bochner's theorem on $\mathbb{R}^m$, $\tilde{F}(x) = \int e^{i\xi\cdot x} \, d\kappa(\xi)$ for a finite positive measure κ. Then $j_*\kappa$ and λ_v have the same Fourier coefficients on Γ, so they are equal, and $\lambda_v(C^c) = 0$. Since v lies in the range of $\mathcal{E}(C)$ iff $\lambda_v(C^c) = 0$, the identification follows.

Factor. Koopman operators commute with complex conjugation and satisfy $T_z(fg) = T_z f \, T_z g$. For bounded $v, w \in H^\Gamma$, $\|T_z(vw) - vw\| \le \|v\|_\infty \|T_z w - w\| + \|w\|_\infty \|T_z v - v\|$, so $vw \in H^\Gamma$, and $\bar{v} \in H^\Gamma$. If $\Phi : \mathbb{C} \to \mathbb{C}$ is Lipschitz, $\|T_z \Phi(v) - \Phi(v)\| = \|\Phi(T_z v) - \Phi(v)\| \le \mathrm{Lip}(\Phi) \|T_z v - v\|$. Applying this to the radial retractions Φ_N onto the disk of radius N shows that the bounded truncations $\Phi_N(v)$ of $v \in H^\Gamma$ lie in H^Γ, and $\Phi_N(v) \to v$ in L^2. Let $\mathcal{G}$ be the completion of the σ-algebra generated by the bounded elements of H^Γ. The bounded elements form a unital $*$-algebra $\mathcal{A}$. For finitely many $v_1, \dots, v_r \in \mathcal{A}$, polynomials in the $v_a, \bar{v}_a$ approximate, uniformly on the compact range of $(v_1, \dots, v_r)$, any continuous function of $(v_1, \dots, v_r)$ (Stone–Weierstrass). Continuous functions are dense in L^2 of the joint distribution, so every L^2 function of $(v_1, \dots, v_r)$ lies in the L^2-closure of $\mathcal{A}$. By the monotone class theorem, such finite-coordinate functions are dense in $L^2(\mathcal{G})$. Since H^Γ is closed, $L^2(\mathcal{G}) \subset H^\Gamma$. Conversely each $v \in H^\Gamma$ is the L^2-limit of the $\mathcal{G}$-measurable $\Phi_N(v)$, so $H^\Gamma \subset L^2(\mathcal{G})$. Orthogonal projection onto $L^2(\mathcal{G})$ is conditional expectation.

Extension. For $x \in \mathbb{R}^m$ choose $z_a \in \Gamma$ with $z_a \to x$. For $v \in H^\Gamma$, $\|T_{z_a} v - T_{z_b} v\| = \|T_{z_a - z_b} v - v\| \to 0$. Let $T_x v$ be the limit. It does not depend on the sequence (interleave two sequences). The group law, unitarity and strong continuity pass to strong limits: e.g. $\|T_x v - T_{x'} v\| = \lim \|T_{z_a - z'_a} v - v\|$, which is small when $x - x'$ is small, uniformly in the choice of approximants. $T_x v$ is the limit of the nonnegative functions $T_{z_a} v$ if $v \ge 0$, and the cone is L^2-closed. Also $T_x 1 = 1$. □

4. The invariant law and the twisted push-forwards

From now on assume that $\mathbb{R}^n$ ($n \ge 3$) has a proper k-coloring c_0.

4.1. **The invariant law.** Let $\Omega \subset [k]^V$ be the set of $\omega : V \to [k]$ with $\omega(z) \ne \omega(z')$ whenever $z, z' \in V$, $|z - z'| = 1$. With the product topology, $[k]^V$ is compact metrizable (V is countable) and Ω is closed, being an intersection of conditions on two coordinates. It is nonempty, since $c_0|_V \in \Omega$. Define

$$(\tau_z \omega)(x) = \omega(x + z), \qquad (\rho_u \omega)(x) = \omega(ux) \qquad (z \in V, \ u \in K).$$

These are homeomorphisms of Ω, since translations and rotations preserve distances in V. They generate a continuous action of $G = V \rtimes K$, which is amenable as an extension of the abelian group K by the abelian group V. Let μ be a G-invariant Borel probability measure on Ω (Section 2). Write $T_z f = f \circ \tau_z$ and $R_u f = f \circ \rho_u$, unitary operators on $L^2(\mu)$. A direct check gives $\tau_{z'} \circ \tau_z = \tau_{z+z'}$, so $T_z T_{z'} = T_{z+z'}$, and $\tau_z \circ \rho_u = \rho_u \circ \tau_{uz}$, so

$$R_u T_z = T_{uz} R_u. \tag{3}$$

Let $f_i = \mathbf{1}\{\omega : \omega(0) = i\}$. Then $T_z f_i = \mathbf{1}\{\omega(z) = i\}$, $\sum_i f_i = 1$, and $R_u f_i = f_i$ since $u0 = 0$. Since every $\omega \in \Omega$ is proper on V,

$$\langle T_z f_i, f_i\rangle = \mu\{\omega : \omega(z) = \omega(0) = i\} = 0 \qquad (z \in V,\ |z| = 1). \tag{4}$$

Fix i and let $\lambda = \lambda_{f_i}$ be the spectral measure of f_i on D_n for the representation $z \mapsto T_z$ of V.

Lemma 4.1. *λ is K-invariant: $(u\cdot)_*\lambda = \lambda$ for all $u \in K$.*

Proof. By (3) and $R_u f_i = f_i$, $\langle T_{uz} f_i, f_i\rangle = \langle R_u T_z R_u^{-1} f_i, f_i\rangle = \langle T_z f_i, R_u^{-1} f_i\rangle = \langle T_z f_i, f_i\rangle$. The Fourier coefficient of $(u\cdot)_*\lambda$ at z is $\int (u \cdot d)(z)\, d\lambda = \int d(uz)\, d\lambda = \langle T_{uz} f_i, f_i\rangle = \langle T_z f_i, f_i\rangle$. Fourier uniqueness gives the claim. □

4.2. **The black box.**

Theorem 4.2 (OpenAI [12, Theorem 2.3])**.** *Every K-invariant Borel probability measure ν on D_P with $\nu(C_P) = 0$ is the normalized Haar measure m_{D_P}.*

Lemma 4.3. *Let ρ be a K-invariant complex Borel measure on D_P with $|\rho|(C_P) = 0$. Then $\rho = c\, m_{D_P}$ for some $c \in \mathbb{C}$. In particular $\int \psi_w\, d\rho = 0$ for $w \in P \setminus \{0\}$.*

Proof. Write $\rho = \rho_1 - \rho_2 + i(\rho_3 - \rho_4)$, where $\rho_1 - \rho_2$ and $\rho_3 - \rho_4$ are the Jordan decompositions of $\operatorname{Re}\rho$ and $\operatorname{Im}\rho$. Let $\alpha_u(e) = u \cdot e$, a Borel automorphism of D_P. Since $(\alpha_u)_*\rho = \rho$ and $(\alpha_u)_*$ commutes with taking real and imaginary parts, $\operatorname{Re}\rho$ and $\operatorname{Im}\rho$ are invariant. Moreover $(\alpha_u)_*\rho_1$ and $(\alpha_u)_*\rho_2$ are mutually singular positive measures with difference $\operatorname{Re}\rho$. By uniqueness of the Jordan decomposition they equal ρ_1 and ρ_2; similarly for ρ_3, ρ_4. Each ρ_a is dominated by $|\rho|$, hence vanishes on C_P. If $\rho_a \neq 0$, Theorem 4.2 applied to $\rho_a/\rho_a(D_P)$ gives $\rho_a = \rho_a(D_P) m_{D_P}$. The last claim holds because $\int \psi_w\, dm_{D_P} = 0$ for $w \neq 0$ (Haar measure is invariant under translation by any e with $e(w) \neq 1$, and such e exist since characters separate points). □

4.3. **Twisted push-forwards.** For $y \in W$ define a complex Borel measure ρ_y on D_P by

$$\int g\, d\rho_y = \int_{D_n} g(\pi d)\, d(0, y)\, d\lambda(d) \qquad (g \text{ bounded Borel on } D_P),$$

i.e. $\rho_y = \pi_*(\psi_{(0,y)}\lambda)$. Then $|\rho_y| \le \pi_*\lambda$.

Lemma 4.4. *ρ_y is K-invariant, and so is its restriction $\rho_y|_{C_P^c}$.*

Proof. For $u \in K$ and bounded Borel g, by (2) and Lemma 4.1,

$$\int g(u \cdot e)\, d\rho_y(e) = \int g(\pi(u \cdot d))\, (u \cdot d)(0, y)\, d\lambda(d) = \int g(\pi d)\, d(0, y)\, d\lambda(d) = \int g\, d\rho_y.$$

Since C_P is K-invariant, the restriction is also invariant. □

Proposition 4.5. *For all $w \in P \setminus \{0\}$ and $y \in W$,*

$$\int_{D_n \setminus A_P} d(w, y)\, d\lambda(d) = 0.$$

Proof. Since $d(w, y) = d(w, 0)d(0, y) = (\pi d)(w)\, d(0, y)$ and $\mathbf{1}_{D_n \setminus A_P} = \mathbf{1}_{C_P^c} \circ \pi$, the integral equals $\int_{C_P^c} \psi_w\, d\rho_y$. The measure $\rho_y|_{C_P^c}$ is K-invariant (Lemma 4.4) and its total variation vanishes on C_P. Lemma 4.3 gives the claim. □

Remark 4.6. Haar measure gives C_P measure zero, so the conclusion of Lemma 4.3 is consistent with the hypothesis $|\rho|(C_P) = 0$. (Proposition 4.5 does not use this.) Suppose $m_{D_P}(C_P) > 0$. C_P is a σ-compact subgroup, so by Steinhaus' theorem $C_P = C_P - C_P$ contains a neighbourhood of 0. Then C_P contains a basic neighbourhood $\{e : |e(z_a) - 1| < \epsilon,\ a \le r\}$, and we may assume

the $z_a \in P$ span $\mathbb{R}^2$. Let $\Lambda = \sum_a \mathbb{Z} z_a$. Every character in $\Lambda^\perp$ lies in that neighbourhood, hence in C_P. But $j(\xi) \in \Lambda^\perp$ forces $\xi \cdot z_a \in 2\pi\mathbb{Z}$ for all a, so ξ lies in a lattice, which is countable. On the other hand, $\Lambda^\perp \cong \widehat{P/\Lambda}$ is an infinite compact group (P/Λ is infinite), hence uncountable. This is a contradiction.

Remark 4.7. The twist $d(0,y)$ is essential. The untwisted push-forward $\pi_*\lambda$ only controls the correlations at $(w,0)$, i.e. at unit vectors in P, which is far from enough. The identity $(u \cdot d)(0,y) = d(0,y)$ makes the twisted push-forward K-invariant for every y, because K fixes W pointwise.

4.4. **The P-continuous part.** Apply Lemma 3.1 to the restricted action of $\Gamma = P \cong \mathbb{F}^2$ (which is literally [12, Lemma 4.1]). Let

$$H_P = \{v \in L^2(\mu) : \ \|T_w v - v\| \to 0 \text{ as } P \ni w \to 0\}.$$

By Lemmas 3.1 and 2.2, H_P is the range of $\mathcal{E}_n(A_P)$, where $\mathcal{E}_n$ is the projection-valued measure of the V-representation. Hence the orthogonal projection Q_P onto H_P commutes with every T_z, $z \in V$. By Lemma 3.1, $H_P = L^2(\mathcal{G}_P)$ for a complete sub-σ-algebra $\mathcal{G}_P$. The P-action on H_P extends to a strongly continuous unitary representation $(T_{(x,0)})_{x\in\mathbb{R}^2}$ preserving positivity. Since $T_{(0,y)}$ ($y \in W$) commutes with all $T_{(w,0)}$, $w \in P$, it commutes with their strong limits $T_{(x,0)}$. Hence

$$T_{(x,y)} := T_{(x,0)} T_{(0,y)} \qquad (x \in \mathbb{R}^2,\ y \in W)$$

is a unitary representation of $\mathbb{R}^2 \times W$ on H_P, strongly continuous in x for fixed y, extending $z \mapsto T_z|_{H_P}$.

Let $q_i = Q_P f_i$. Then $\lambda_{q_i} = \mathbf{1}_{A_P}\lambda$, so for $z \in V$,

$$\langle T_z f_i, f_i\rangle = \langle T_z q_i, q_i\rangle + \int_{D_n\setminus A_P} d(z)\, d\lambda(d).$$

Combining with (4) and Proposition 4.5:

$$\langle T_{(w,y)} q_i, q_i\rangle = 0 \qquad \big((w,y) \in V,\ w \neq 0,\ |w|^2 + |y|^2 = 1\big). \tag{5}$$

Lemma 4.8. *For every $y \in W$ with $|y| < 1$, put $r(y) = \sqrt{1-|y|^2}$. Then $\langle T_{(x,y)} q_i, q_i\rangle = 0$ for all $x \in \mathbb{R}^2$ with $|x| = r(y)$.*

Proof. Since $|y|^2 \in \mathbb{F}$ and $\mathbb{F}$ is real closed, $r(y) \in \mathbb{F}$ and $r(y) > 0$. The set $S^1_{\mathbb{F}} = S^1 \cap \mathbb{F}^2$ is dense in S^1; it contains the rational points $\big(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\big)$, $t \in \mathbb{Q}$. Hence $\{w \in \mathbb{F}^2 : |w| = r(y)\} = r(y) S^1_{\mathbb{F}}$ is dense in the circle of radius $r(y)$. Each such w is nonzero and satisfies $|w|^2 + |y|^2 = 1$, so (5) applies. The function $x \mapsto \langle T_{(x,0)} T_{(0,y)} q_i, q_i\rangle$ is continuous on $\mathbb{R}^2$, which gives the claim. □

5. Removing spectrum that is wild along W

5.1. **The fully continuous part.** Apply Lemma 3.1 to the action of $\Gamma = V = \mathbb{F}^n$:

$$H_{\mathrm{c}} = \{v : \ \|T_z v - v\| \to 0 \text{ as } V \ni z \to 0\} = \operatorname{ran} \mathcal{E}_n(C_n) = L^2(\mathcal{G}_{\mathrm{c}}),$$

and the V-action on H_{c} extends to a strongly continuous, positivity-preserving unitary representation $(T_x)_{x\in\mathbb{R}^n}$. Clearly $H_{\mathrm{c}} \subset H_P$, consistently with $C_n \subset A_P$. On H_{c}, the operators $T_{(x,y)}$ ($x \in \mathbb{R}^2$, $y \in W$) defined in Section 4 coincide with the restrictions of the $\mathbb{R}^n$-extension, since both are strong limits of $T_{(w,y)}$ as $P \ni w \to x$. Let

$$p_i = \mathbb{E}(f_i \mid \mathcal{G}_{\mathrm{c}}), \qquad b_i = q_i - p_i.$$

Since $H_{\mathrm{c}} \subset H_P$, the projection Q_{c} onto H_{c} satisfies $Q_{\mathrm{c}} Q_P = Q_{\mathrm{c}}$, so $p_i = Q_{\mathrm{c}} f_i = Q_{\mathrm{c}} q_i$. Thus $b_i \in H_{\mathrm{b}} := H_P \ominus H_{\mathrm{c}}$. Conditional expectation gives $0 \le p_i \le 1$ and $\sum_i p_i = 1$ a.e.

Lemma 5.1. *H_{c} and H_{b} are invariant under $T_{(x,y)}$ for all $x \in \mathbb{R}^2$, $y \in W$.*

Proof. H_{c} is invariant under every T_z, $z \in V$ (Lemma 3.1), and it is closed, so it is invariant under the strong limits $T_{(x,y)}$. Each $T_{(x,y)}$ is unitary on H_P with inverse $T_{(-x,-y)}$, which also preserves H_{c}. Hence it preserves $H_P \ominus H_{\mathrm{c}}$. □

Lemma 5.2. *For $y \in W$ with $|y| < 1$ and $x \in \mathbb{R}^2$ with $|x| = r(y)$,*

$$\langle T_{(x,y)} b_i, b_i \rangle = -\langle T_{(x,y)} p_i, p_i \rangle.$$

Proof. Expand $\langle T_{(x,y)}(p_i + b_i), p_i + b_i \rangle$. The cross terms vanish by Lemma 5.1: $T_{(x,y)} p_i \in H_{\mathrm{c}} \perp b_i$ and $T_{(x,y)} b_i \in H_{\mathrm{b}} \perp p_i$. The left side vanishes by Lemma 4.8. □

5.2. **Averaged correlations.** Fix i and write $p = p_i$, $b = b_i$. Let $\bar{B} = \{y \in \mathbb{R}^{n-2} : |y| \le 1\}$, B° its interior, and $r(y) = \sqrt{1 - |y|^2}$ for $y \in \bar{B}$. Let $d\theta$ be normalized arc length on $S^1 \subset \mathbb{R}^2$. Define

$$h(y) = \int_{S^1} \langle T_{(r(y)\theta,\, y)} p, p \rangle \, d\theta \qquad (y \in \bar{B}),$$

using the $\mathbb{R}^n$-action on H_{c}, and

$$g(y) = \int_{S^1} \langle T_{(r(y)\theta,\, y)} b, b \rangle \, d\theta \qquad (y \in W \cap B^\circ).$$

Also define, for $y \in \bar{B}$,

$$\beta(y) = \int_{S^1} T_{(-r(y)\theta,\, 0)} b \, d\theta \ \in H_{\mathrm{b}},$$

the integral of a norm-continuous H_{b}-valued function on S^1 (Riemann/Bochner integral). It lies in H_{b} by Lemma 5.1 and closedness.

Lemma 5.3. *(a) h is continuous and nonnegative on $\bar{B}$. For each $y \in \bar{B}$, the integrand $\theta \mapsto \langle T_{(r(y)\theta,y)} p, p \rangle$ is continuous and nonnegative.*
(b) $g(y) = \langle T_{(0,y)} b, \beta(y) \rangle$ and $g(y) = -h(y)$ for $y \in W \cap B^\circ$. Moreover $|g| \le \|b\|^2$.
(c) $\beta : \bar{B} \to H_{\mathrm{b}}$ is norm-continuous.

Proof. (a) $(x, y) \mapsto T_{(x,y)} p$ is norm-continuous on $\mathbb{R}^n$, and r is continuous, so the integrand is jointly continuous in $(\theta, y) \in S^1 \times \bar{B}$. Integration over the compact S^1 preserves continuity. Nonnegativity: $T_x p \ge 0$ and $p \ge 0$ pointwise a.e., so $\langle T_x p, p \rangle = \int (T_x p)\, p \, d\mu \ge 0$.

(b) By unitarity and the group law on H_P, $\langle T_{(r\theta,y)} b, b \rangle = \langle T_{(0,y)} b, T_{(-r\theta,0)} b \rangle$. Integrating in θ and moving the integral inside the inner product (continuity of the inner product and real weights) gives the formula. Lemma 5.2 gives $g = -h$ on $W \cap B^\circ$. The bound is Cauchy–Schwarz.

(c) For $y, y' \in \bar{B}$, $\|\beta(y) - \beta(y')\| \le \sup_\theta \|T_{(-r(y)\theta,0)} b - T_{(-r(y')\theta,0)} b\| = \sup_\theta \|T_{((r(y)-r(y'))\theta,0)} b - b\|$. This tends to 0 as $y' \to y$, because $\|T_{(x,0)} b - b\| \to 0$ as $x \to 0$ in $\mathbb{R}^2$ and $|(r(y) - r(y'))\theta| = |r(y) - r(y')|$. □

5.3. **An invariant mean on W.** $\mathbb{Z}^{n-2}$ is a subgroup of $\mathbb{F}^{n-2} \cong W$. Let m be a translation-invariant mean on $\ell^\infty(W/\mathbb{Z}^{n-2})$, which exists because the discrete group $W/\mathbb{Z}^{n-2}$ is abelian, hence amenable [2, Theorem G.2.1]. Call $\phi : W \to \mathbb{C}$ *admissible* if it is bounded and vanishes outside a bounded set. For admissible ϕ define the periodization $\mathcal{P}\phi(t) = \sum_{k \in \mathbb{Z}^{n-2}} \phi(t + k)$. If ϕ vanishes outside $[-R, R]^{n-2}$, at most $(2R + 2)^{n-2}$ terms are nonzero for each t, so $\mathcal{P}\phi$ is a bounded $\mathbb{Z}^{n-2}$-periodic function on W, i.e. an element of $\ell^\infty(W/\mathbb{Z}^{n-2})$. Put

$$I(\phi) = \mathsf{m}(\mathcal{P}\phi).$$

This is the $(n-2)$-dimensional analogue of the line average [12, (2.3)].

Lemma 5.4. *I is linear and positive, satisfies $|I(\phi)| \le I(|\phi|)$, and is W-invariant: $I(\phi(\cdot+s)) = I(\phi)$ for $s \in W$. For $\phi \in C_{\mathrm{c}}(\mathbb{R}^{n-2})$, $I(\phi|_W) = \int_{\mathbb{R}^{n-2}} \phi(t)\,dt$.*

Proof. Linearity and positivity follow from those of $\mathcal{P}$ and m. For complex ϕ choose α with $e^{i\alpha}I(\phi) = |I(\phi)|$. Since I maps real functions to real numbers (by positivity), $|I(\phi)| = I(e^{i\alpha}\phi) = I(\mathrm{Re}(e^{i\alpha}\phi)) + iI(\mathrm{Im}(e^{i\alpha}\phi))$ has vanishing imaginary part, so $|I(\phi)| = I(\mathrm{Re}(e^{i\alpha}\phi)) \le I(|\phi|)$. Invariance: $\mathcal{P}(\phi(\cdot+s)) = (\mathcal{P}\phi)(\cdot+s)$, and m is invariant.

Let $\phi \in C_{\mathrm{c}}(\mathbb{R}^{n-2})$ and $\Phi(t) = \sum_k \phi(t+k)$ for $t \in \mathbb{R}^{n-2}$. Then Φ is continuous, $\mathbb{Z}^{n-2}$-periodic, and $\mathcal{P}(\phi|_W) = \Phi|_W$. Define $L : C(\mathbb{T}^{n-2}) \to \mathbb{C}$, $\mathbb{T}^{n-2} = \mathbb{R}^{n-2}/\mathbb{Z}^{n-2}$, by $L(\Psi) = \mathsf{m}(\Psi|_W)$, viewing Ψ as a periodic function on $\mathbb{R}^{n-2}$. Then L is positive, linear, $L(1) = 1$, $|L(\Psi)| \le \sup|\Psi|$, and $L(\Psi(\cdot+s)) = L(\Psi)$ for $s \in W$. Let $e_\ell(t) = e^{2\pi i \ell\cdot t}$ with $\ell \in \mathbb{Z}^{n-2} \setminus \{0\}$. Choose a coordinate a with $\ell_a \neq 0$ and let $s = (2\ell_a)^{-1}\mathbf{e}_a \in \mathbb{Q}^{n-2} \subset W$. Then $e_\ell(t+s) = -e_\ell(t)$, so $L(e_\ell) = -L(e_\ell) = 0$. Thus L agrees with the Haar integral on trigonometric polynomials, which are uniformly dense in $C(\mathbb{T}^{n-2})$ (Stone–Weierstrass), and both functionals are bounded, so $L(\Psi) = \int_{[0,1)^{n-2}} \Psi$. Hence $I(\phi|_W) = \int_{[0,1)^{n-2}} \sum_k \phi(t+k)\,dt = \int_{\mathbb{R}^{n-2}} \phi$. □

Weak vector integrals. Let $\mathcal{V} : W \to H$ be bounded with bounded support, H a Hilbert space. The map $\eta \mapsto I(\langle \mathcal{V}(\cdot), \eta\rangle)$ is conjugate-linear, and by Lemma 5.4 it is bounded by $I(\|\mathcal{V}(\cdot)\|)\|\eta\|$. By the Riesz representation theorem there is a unique $I(\mathcal{V}) \in H$ with

$$\langle I(\mathcal{V}), \eta\rangle = I\big(\langle \mathcal{V}(\cdot), \eta\rangle\big) \quad (\eta \in H), \qquad \|I(\mathcal{V})\| \le I(\|\mathcal{V}(\cdot)\|).$$

No measurability of $\mathcal{V}$ is needed, since W is discrete and m is defined on all bounded functions.

Lemma 5.5. *For $\psi \in C_{\mathrm{c}}(\mathbb{R}^{n-2})$ let $V_\psi = I\big(y \mapsto \psi(y)\,T_{(0,y)}b\big)$, $y \in W$. Then $V_\psi = 0$.*

Proof. $V_\psi \in H_{\mathrm{b}}$. If $\eta \perp H_{\mathrm{b}}$, then $\langle T_{(0,y)}b, \eta\rangle = 0$ for all $y \in W$ (Lemma 5.1), so $\langle V_\psi, \eta\rangle = 0$. Hence $V_\psi \in (H_{\mathrm{b}}^{\perp})^{\perp} = H_{\mathrm{b}}$.

Continuity along W. For $s \in W$ and $\eta \in L^2$,

$$\langle T_{(0,s)}V_\psi, \eta\rangle = \langle V_\psi, T_{(0,-s)}\eta\rangle = I\big(\psi(y)\langle T_{(0,y+s)}b, \eta\rangle\big) = I\big(\psi(y-s)\langle T_{(0,y)}b, \eta\rangle\big),$$

by W-invariance of I. Hence $T_{(0,s)}V_\psi - V_\psi = I\big((\psi(y-s)-\psi(y))T_{(0,y)}b\big)$, and by Lemma 5.4

$$\|T_{(0,s)}V_\psi - V_\psi\| \le \|b\|\, I\big(|\psi(\cdot - s) - \psi|\big) = \|b\| \int_{\mathbb{R}^{n-2}} |\psi(t-s) - \psi(t)|\,dt \xrightarrow[s\to 0]{} 0,$$

by uniform continuity and compact support of ψ (the function $|\psi(\cdot - s) - \psi|$ lies in C_{c}).

Continuity along V. Since $V_\psi \in H_{\mathrm{b}} \subset H_P$, $\|T_{(w,0)}V_\psi - V_\psi\| \to 0$ as $P \ni w \to 0$. For $(w,s) \in V$,

$$\|T_{(w,s)}V_\psi - V_\psi\| \le \|T_{(w,0)}(T_{(0,s)}V_\psi - V_\psi)\| + \|T_{(w,0)}V_\psi - V_\psi\| = \|T_{(0,s)}V_\psi - V_\psi\| + \|T_{(w,0)}V_\psi - V_\psi\| \to 0.$$

So $V_\psi \in H_{\mathrm{c}} \cap H_{\mathrm{b}} = \{0\}$. □

Proposition 5.6. *For every $\phi \in C_{\mathrm{c}}(B^\circ)$, the function ϕg on W (set to 0 outside B°) is admissible and $I(\phi g) = 0$.*

Proof. Admissibility follows from $|g| \le \|b\|^2$. Fix $\epsilon > 0$ and let $K_0 = \operatorname{supp}\phi \subset B^\circ$, which is compact. By Lemma 5.3(c), β is uniformly continuous on the compact set $\bar{B}$. Choose $\delta > 0$ with $\|\beta(y) - \beta(y')\| < \epsilon$ whenever $y, y' \in \bar{B}$, $|y - y'| < \delta$, and with $\delta < \operatorname{dist}(K_0, \partial B)$. Cover K_0 by finitely many balls $B(y_a, \delta)$ with $y_a \in K_0$, $a = 1, \dots, N$. Let $\chi_1, \dots, \chi_N$ be a continuous partition of unity on K_0 subordinate to this cover: $\chi_a \in C_{\mathrm{c}}(B(y_a,\delta))$, $\chi_a \ge 0$, $\sum_a \chi_a = 1$ on K_0. For $y \in K_0$,

$$\Big\|\beta(y) - \sum_a \chi_a(y)\beta(y_a)\Big\| \le \sum_a \chi_a(y)\|\beta(y) - \beta(y_a)\| < \epsilon.$$

By Lemma 5.3(b), for $y \in W$, $\phi(y)g(y) = \phi(y)\langle T_{(0,y)}b, \beta(y)\rangle$, so

$$\phi(y)g(y) = \sum_a \phi(y)\chi_a(y)\langle T_{(0,y)}b, \beta(y_a)\rangle + \phi(y)\Big\langle T_{(0,y)}b,\ \beta(y) - \sum_a \chi_a(y)\beta(y_a)\Big\rangle.$$

Apply I. The a-th term of the sum gives $\langle V_{\phi\chi_a}, \beta(y_a)\rangle = 0$ by Lemma 5.5, since $\phi\chi_a \in C_{\mathrm{c}}(\mathbb{R}^{n-2})$. The last term is bounded in modulus by $\epsilon\|b\|\,|\phi(y)|$, so by Lemma 5.4 its I-value is at most $\epsilon\|b\|\,I(|\phi|) = \epsilon\|b\|\int|\phi|$. Hence $|I(\phi g)| \le \epsilon\|b\|\int|\phi|$ for every $\epsilon > 0$. □

Proposition 5.7. *For each i and every $u \in S^{n-1}$, $\langle T_u p_i, p_i\rangle = 0$. Consequently $\langle T_x p_i, T_{x+u}p_i\rangle = 0$ for all $x \in \mathbb{R}^n$, $u \in S^{n-1}$.*

Proof. Let $\phi \in C_{\mathrm{c}}(B^\circ)$. Then $\phi h \in C_{\mathrm{c}}(\mathbb{R}^{n-2})$ (Lemma 5.3(a)), and on W we have $\phi h = -\phi g$ (Lemma 5.3(b)). By Lemma 5.4 and Proposition 5.6,

$$\int_{\mathbb{R}^{n-2}} \phi h = I\big((\phi h)|_W\big) = -I(\phi g) = 0.$$

Since h is continuous on B°, $h \equiv 0$ there. For $y \in B^\circ$, the integrand $\theta \mapsto \langle T_{(r(y)\theta,y)}p, p\rangle$ is continuous, nonnegative, and has integral 0, so it vanishes identically. Thus $\langle T_u p, p\rangle = 0$ for every $u = (u', y) \in S^{n-1}$ with $|y| < 1$, i.e. with $u' \neq 0$. These u are dense in S^{n-1}, and $u \mapsto \langle T_u p, p\rangle$ is continuous, which proves the first claim. The second follows from $\langle T_x p, T_{x+u}p\rangle = \langle T_x p, T_x T_u p\rangle = \langle p, T_u p\rangle = \overline{\langle T_u p, p\rangle}$. □

Remark 5.8. The argument never shows that the correlations $\langle T_z b_i, b_i\rangle$ of the remainder vanish at individual unit vectors z with nonzero W-component. It shows only that their circle averages vanish after averaging over W with I. The positivity of the continuous part then turns this averaged statement into a pointwise one for p_i, which is all we need.

Remark 5.9. For $n = 2$, $W = 0$, $A_P = C_n$, $H_P = H_{\mathrm{c}}$, $b_i = 0$, and Lemma 4.8 with $y = 0$ is the statement $\langle T_u p_i, p_i\rangle = 0$, $u \in S^1$, of [12, §4.1]. Section 5 is vacuous, and the proof reduces to OpenAI's.

Remark 5.10 (Why one plane suffices). Averaging over all of $SO(n, \mathbb{F})$ is impossible in general: for $n \ge 3$ this group contains a free subgroup of rank two (take two rotations of $\mathbb{R}^3$ with algebraic entries generating a free group, as in the Banach–Tarski paradox), so it is not amenable. Our proof uses rotation invariance only inside P, uniformly over all offsets $y \in W$ thanks to the twist. In the W-directions it uses only translations (an abelian invariant mean) and positivity of the continuous part.

6. The measurable sample and the proof of Theorem 1.2

Proposition 6.1. *If $\mathbb{R}^n$ has a proper k-coloring, it has a weak measurable k-coloring.*

Proof. For $n = 2$ this is [12, Proposition 4.2]; let $n \ge 3$ and let $p_1, \dots, p_k$ be as above. We follow [12, §4.1].

Jointly measurable fields. Fix i. The map $x \mapsto T_x p_i$ from $\mathbb{R}^n$ to $L^2(\mu)$ is continuous, and $L^2(\mu)$ is separable (Ω is compact metrizable). On the ball $B(0, M)$ there are Borel simple maps $s_N : B(0, M) \to L^2(\mu)$, each taking finitely many values on Borel pieces, with $\int_{B(0,M)} \|s_N(x) - T_x p_i\|^2 dx \le 4^{-N}$. (Partition $B(0, M)$ into Borel sets of small diameter, use uniform continuity, and take one value per piece.) Choose Borel representatives of the finitely many values. This gives jointly Borel functions $S_N(x, \omega)$ on $B(0, M) \times \Omega$. Then $\sum_N \|S_{N+1} - S_N\|_{L^2(dx\otimes\mu)} < \infty$, so S_N converges in $L^2(dx \otimes \mu)$ and $dx \otimes \mu$-a.e. to a Borel limit S. Moreover $\int \|S_N(x, \cdot) - T_x p_i\|^2 dx$ is summable, so for a.e. x we have $S_N(x, \cdot) \to T_x p_i$ in $L^2(\mu)$, hence $S(x, \cdot) = T_x p_i$ μ-a.e. Doing this on the disjoint annuli $\{M - 1 \le |x| < M\}$ produces a Borel field $\tilde{H}_i$ on $\mathbb{R}^n \times \Omega$ such that $\tilde{H}_i(x, \cdot)$

represents $T_x p_i$ for a.e. x. Since $0 \le T_x p_i \le 1$ (positivity applied to p_i and to $1-p_i$, together with $T_x 1 = 1$), the field $H_i = \min(1, \max(0, \operatorname{Re} \tilde{H}_i))$ has the same property, and $0 \le H_i \le 1$ everywhere.

Exceptional sets. Let $N_0 \subset \mathbb{R}^n$ be a Lebesgue null set outside which $H_i(x,\cdot)$ represents $T_x p_i$ for every i. For each u, $\{x : x+u \in N_0\} = N_0 - u$ is null, so by Fubini the set of $(x,u) \in \mathbb{R}^n \times S^{n-1}$ with $x \in N_0$ or $x+u \in N_0$ is $dx \otimes \sigma$-null. For (x,u) outside it, Proposition 5.7 gives

$$\int_\Omega H_i(x,\omega) H_i(x+u,\omega)\, d\mu(\omega) = \langle T_x p_i, T_{x+u} p_i \rangle = 0.$$

By Tonelli (all integrands are nonnegative and jointly Borel), for every integer $M \ge 1$,

$$\int_\Omega \sum_{i=1}^{k} \int_{B(0,M)} \int_{S^{n-1}} H_i(x,\omega) H_i(x+u,\omega)\, d\sigma(u)\, dx\, d\mu(\omega) = 0.$$

Also, for $x \notin N_0$, $\sum_i H_i(x,\cdot)$ represents $T_x \sum_i p_i = T_x 1 = 1$. So the Borel set $\{(x,\omega) : \sum_i H_i(x,\omega) \ne 1\}$ has μ-null sections for a.e. x, hence it is $dx \otimes \mu$-null, and for μ-a.e. ω its ω-section is Lebesgue-null.

Choosing a sample. The conditions above are countably many. Choose ω such that, for every M, $\sum_i \int_{B(0,M)} \int H_i(x,\omega) H_i(x+u,\omega)\, d\sigma\, dx = 0$, and such that $Z = \{x : \sum_i H_i(x,\omega) \ne 1\}$ is null. Define $c(x)$ to be the least i with $H_i(x,\omega) > 0$ if $x \notin Z$, and $c(x) = 1$ on Z. Then c is Borel. If $c(x) = c(x+u) = i$ with $x, x+u \notin Z$, then $H_i(x,\omega) H_i(x+u,\omega) > 0$, and this happens only on a $dx \otimes \sigma$-null subset of $B(0,M) \times S^{n-1}$. Pairs with $x \in Z$ or $x+u \in Z$ form a null set by the Fubini argument above. Since every R is at most some M, c is a weak measurable k-coloring. □

6.1. **The converse.** For a measurable $A \subset \mathbb{R}^n$ let

$$\operatorname{Dens}(A) = \Big\{ x \in \mathbb{R}^n : \lim_{h \downarrow 0} \frac{|A \cap B(x,h)|}{|B(x,h)|} = 1 \Big\}.$$

Since $(x,h) \mapsto |A \cap B(x,h)|/|B(x,h)|$ is continuous on $\mathbb{R}^n \times (0,\infty)$, the limit can be taken along rational h, so $\operatorname{Dens}(A)$ is Borel. By the Lebesgue density theorem, $|A \triangle \operatorname{Dens}(A)| = 0$.

Lemma 6.2 (cf. [12, Lemma 4.3])**.** *Let c be a weak measurable k-coloring of $\mathbb{R}^n$ with classes $A_1, \dots, A_k$, and let $D_i = \operatorname{Dens}(A_i)$. Then no D_i contains two points at distance 1, the D_i are pairwise disjoint Borel sets, and $\mathbb{R}^n \setminus \bigcup_i D_i$ is null.*

Proof. Suppose $x, y = x+u \in D_i$ with $|u| = 1$. For $0 < h < 1$ let $\Theta_h = \{u' \in S^{n-1} : |u'-u| < h\}$, a spherical cap with $\sigma(\Theta_h) > 0$. Let x' be uniform in $B(x,h)$ and, independently, u' uniform on Θ_h (w.r.t. σ). Then $\Pr(x' \notin A_i) = |B(x,h) \setminus A_i|/|B(x,h)| \to 0$. Conditionally on u', the point $x'+u'$ is uniform on $B(x+u',h)$, so the law of $x'+u'$ has density at most $1/|B(0,h)|$. It is supported in $B(y,2h)$, since $|x'+u'-y| \le |x'-x| + |u'-u| < 2h$. Hence

$$\Pr(x'+u' \notin A_i) \le \frac{|B(y,2h) \setminus A_i|}{|B(0,h)|} = 2^n \frac{|B(y,2h) \setminus A_i|}{|B(y,2h)|} \to 0.$$

For small h, therefore, $\Pr(x' \in A_i,\ x'+u' \in A_i) > 0$. The law of (x',u') has density $(|B(0,h)|\sigma(\Theta_h))^{-1} \mathbf{1}_{B(x,h)\times\Theta_h}$ with respect to $dx \otimes \sigma$. Taking $R > |x| + 1$,

$$\int_{B(0,R)} \int_{S^{n-1}} \mathbf{1}_{A_i}(x') \mathbf{1}_{A_i}(x'+u')\, d\sigma(u')\, dx' \ \ge\ |B(0,h)|\, \sigma(\Theta_h)\, \Pr(x' \in A_i,\ x'+u' \in A_i) > 0,$$

contradicting Definition 1.1. If $x \in D_i \cap D_j$ with $i \ne j$, then the densities of A_i and A_j at x would both be 1, while $A_i \cap A_j = \emptyset$ forces their sum to be at most 1. Finally $\bigcup_i A_i = \mathbb{R}^n$ and $|A_i \setminus D_i| = 0$. □

Proof of Theorem 1.2. The forward implication is Proposition 6.1. Conversely, let c be a weak measurable k-coloring and let $D = \bigcup_i D_i$ as in Lemma 6.2. Let $X \subset \mathbb{R}^n$ be finite. The set $\bigcap_{x \in X}(D - x)$ is a finite intersection of conull sets, hence nonempty. Pick t in it, so that $X + t \subset D$. Coloring $x \in X$ by the index i with $x + t \in D_i$ is a proper k-coloring of the unit-distance graph on X, by Lemma 6.2. By the de Bruijn–Erdős theorem [3, Theorem 1], the unit-distance graph on $\mathbb{R}^n$ is k-colorable. Its proof in ZFC is product compactness: in $[k]^{\mathbb{R}^n}$ each unit-edge condition is closed, and every finite subfamily is satisfiable. □

7. Consequences

Proof of Corollary 1.3. Let $k = \chi(\mathbb{R}^n)$, which is finite (e.g. $\chi(\mathbb{R}^n) \le (3 + o(1))^n$ [10]; any finite bound suffices). By Theorem 1.2 there is a weak measurable k-coloring. By Lemma 6.2 there are pairwise disjoint Borel sets $D_1, \dots, D_k$, none containing two points at distance 1, whose union D is conull. Let $(Q_R)_{R>0}$ be the cubes $[-R, R]^n$, or more generally any fixed exhaustion (e.g. balls $B(0, R)$). For each R, $\sum_i |D_i \cap Q_R| = |Q_R|$, so some $i = i(R)$ has $|D_i \cap Q_R|/|Q_R| \ge 1/k$. Some index i_0 occurs as $i(R)$ for a sequence $R \to \infty$, so $\limsup_R |D_{i_0} \cap Q_R|/|Q_R| \ge 1/k$. Hence $m_1(\mathbb{R}^n) \ge 1/k$ (and the same for the exhaustion chosen), and a fortiori $m_1^*(\mathbb{R}^n) \ge 1/k$. Since k is an integer, $k \ge \lceil 1/m_1(\mathbb{R}^n) \rceil$. □

Remark 7.1. The argument gives slightly more: any lower bound for $\chi_m(\mathbb{R}^n)$ whose proof only uses a measurable *almost everywhere* proper coloring by unit-free sets (equivalently, a weak measurable coloring) transfers to $\chi(\mathbb{R}^n)$. We do not claim $\chi(\mathbb{R}^n) = \chi_m(\mathbb{R}^n)$: a measurable coloring must also color the exceptional null set, and Theorem 1.2 says nothing about that.

7.1. **Numerical inputs.** We list the upper bounds for $m_1(\mathbb{R}^n)$ that we use, with their sources and density conventions.

- **Oliveira–Vallentin** [13, Table 3.1] (centred cubes): $m_1(\mathbb{R}^3) \le 0.165609$, …, $m_1(\mathbb{R}^{24}) \le 0.00047489$. They deduced $\chi_m(\mathbb{R}^3) \ge 7$ in 2010; the previous measurable bound was 6. (Separately, 6 is Nechushtan's bound for the unrestricted $\chi(\mathbb{R}^3)$ [11].) We read the table in the preprint version of [13].
- **Bachoc–Passuello–Thiery** [1] (centred cubes): $m_1(\mathbb{R}^n) \le (1.268 + o(1))^{-n}$ [1, Theorem 1.1], and improved numerical bounds for $4 \le n \le 24$ [1, Table 2], for instance $m_1(\mathbb{R}^4) \le 0.1000625$, $m_1(\mathbb{R}^9) \le 0.0146577$, $m_1(\mathbb{R}^{24}) \le 1.84366 \cdot 10^{-4}$. The $n = 4$ value 0.1000625 is the one in the arXiv version of [1, Table 2]; [6, Table 2] quotes it as 0.1000620. The ceiling (10) is the same either way, and the bound we use for $n = 4$ is DOV's.
- **DeCorte–Oliveira–Vallentin** [6, Table 2] (uniform density $\overline{\delta}^*$; bounds for m_1^*, hence for m_1): $m_1(\mathbb{R}^3) \le 0.1532996$, $m_1(\mathbb{R}^4) \le 0.0985701$, $m_1(\mathbb{R}^5) \le 0.0624485$, $m_1(\mathbb{R}^6) \le 0.0450325$, $m_1(\mathbb{R}^7) \le 0.0260782$, $m_1(\mathbb{R}^8) \le 0.0190945$. These improve [1] for $3 \le n \le 8$, and DOV state the resulting bounds $\chi_m(\mathbb{R}^n) \ge 7, 11, 17, 23, 39, 53$.

For $n = 3$ both $1/0.165609 \approx 6.038$ and $1/0.1532996 \approx 6.523$ exceed 6, so $\chi(\mathbb{R}^3) \ge 7$, improving Nechushtan's $\chi(\mathbb{R}^3) \ge 6$ [11]. The current upper bound is $\chi(\mathbb{R}^3) \le 15$ [5, 15].

For $n \in \{16, 20, 24\}$ we did not locate a reliable recent tabulation of the best lower bounds for $\chi(\mathbb{R}^n)$, so we leave the entries blank. We expect, but have not checked, that known constructions in these dimensions give much smaller values.

7.2. **Asymptotics.** By [1, Theorem 1.1], $m_1(\mathbb{R}^n) \le (1.268 + o(1))^{-n}$. Corollary 1.3 gives

$$\chi(\mathbb{R}^n) \;\ge\; (1.268 + o(1))^n.$$

The previous record for unrestricted colorings, due to Raigorodskii [16], was $(1.239\ldots + o(1))^n$, improving Frankl–Wilson's $(1.207\ldots + o(1))^n$ [8]. The upper bound is $(3 + o(1))^n$ [10].

TABLE 1. Lower bounds $\lceil 1/m_1(\mathbb{R}^n)\rceil$ for $\chi(\mathbb{R}^n)$ (Corollary 1.3) for selected n, compared with earlier lower bounds for $\chi(\mathbb{R}^n)$. The "previous" column gives the largest bound we found, with its source. Entries marked † come from arXiv preprints, and ‡ marks a 2026 preprint claim that we have not verified. The column may be incomplete. The value for $n = 4$ is attributed to Exoo in [4]; we did not locate the original. The m_1 values from [1, Table 2] for $n \geq 9$ were read from the arXiv version and cross-checked only against the χ_m bounds stated there. We did not re-verify them independently.

n	previous lower bound for $\chi(\mathbb{R}^n)$	upper bound for $m_1(\mathbb{R}^n)$	source	$\lceil 1/m_1\rceil$
3	6 [11]	0.1532996	[6]	7
4	9 (Exoo, see [4])	0.0985701	[6]	11
5	9 (see [4])	0.0624485	[6]	17
6	$12^\dagger$ [7]	0.0450325	[6]	23
7	$16^\dagger$ [7]	0.0260782	[6]	39
8	$19^\dagger$ [9]; $25^\ddagger$ [14]	0.0190945	[6]	53
9	21 (see [4], [9])	0.0146577	[1]	69
10	$26^\dagger$ [7, 9]	0.0111621	[1]	90
11	$32^\dagger$ [9]	0.00862918	[1]	116
12	$36^\dagger$ [7]	0.00611112	[1]	164
16	—	0.00161646	[1]	619
20	—	0.000469095	[1]	2132
24	—	0.000184366	[1]	5424

8. Points requiring care

(1) *Dependence on* [12, Theorem 2.3]. Everything rests on the planar rigidity theorem, whose proof occupies [12, §§2–3]. We have not re-verified it.

(2) *Application to complex measures.* Theorem 4.2 is stated for probability measures. We apply it to the four Jordan parts of a K-invariant complex measure (Lemma 4.3). Uniqueness of the Jordan decomposition makes these parts K-invariant, and they vanish on C_P because they are dominated by $|\rho|$.

(3) *Spectral subspaces for the restricted action.* The identification $H_P = \operatorname{ran}\mathcal{E}_n(A_P)$ uses uniqueness of the projection-valued measure (Lemma 2.2). It implies that Q_P commutes with the full V-action, which is needed for H_{b} to be W-invariant.

(4) *Weak integrals and the invariant mean.* I is defined on all bounded, boundedly supported functions on the discrete group W, so no measurability of $y \mapsto T_{(0,y)}b$ in the real topology is needed, as in [12, §2.2]. The only real-variable input is Lemma 5.4, which concerns continuous functions.

(5) *Real versus algebraic parameters.* h and β are defined at real y, while g and the identity $g = -h$ hold only on $W \cap B^\circ$. The proof of Proposition 5.6 evaluates β only at the points y_a. The conclusion passes to real y through $\int \phi h = I(\phi h|_W)$.

(6) *Density conventions.* Corollary 1.3 is proved for the centred upper density (cubes $[-T,T]^n$), and the same argument works for any fixed exhaustion. [13, 1] bound the centred density. [6] bound the uniform density $\overline{\delta}^*$ (supremum over centres), which dominates the centred one, so their bounds also bound m_1. The sets D_i avoid distance one exactly, not just almost everywhere, so they are admissible in every convention.

(7) *Choice.* The forward direction uses an arbitrary proper coloring and an invariant mean; the converse uses compactness of $[k]^{\mathbb{R}^n}$. All of this is within ZFC.

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